

AMAN KUMAR PANDA

Junagarh, Odisha, 766014 | +91-9937552371 | +91-7849052371 | amanpanda987@gmail.com | linkedin.com/in/amanpanda24 | github.com/349100

PROFESSIONAL SUMMARY

Cybersecurity analyst and Master of Computer Applications candidate specializing in offensive security, digital forensics, threat intelligence, and network detection and response. Hands-on experience in vulnerability assessment, penetration testing, network security, and machine learning-based anomaly detection. Developer of PhantomEye, an AI-powered OSINT framework, and NeuroShield NDR. Seeking to contribute technical expertise to security operations, threat intelligence, vulnerability research, or NDR engineering roles.

CORE COMPETENCIES

Network Security | Threat Intelligence | Incident Response | Vulnerability Assessment | Penetration Testing | OSINT | Web Application Security | Network Detection and Response | Digital Forensics | Malware Analysis | Risk Assessment.

TECHNICAL SKILLS

Security Tools: Burp Suite, Nessus, Nmap, Wireshark, Metasploit Framework, OWASP ZAP, OpenVAS, FTK Imager, Maltego, Kali Linux, Parrot OS

Networking: TCP/IP, DNS, HTTP/HTTPS, Packet Analysis, Flow Reassembly, Network Protocols, Subnetting, VLANs.

Frameworks & Standards: OWASP Testing Guide, OWASP Top 10, NIST Cybersecurity Framework, ISO 27001, CVSS Scoring, SDLC

PROFESSIONAL EXPERIENCE

Cybersecurity Intern & Professional Trainee

Hacker School (A Unit of Cartel Software) | 07/2024 – 12/2024

- Completed 200+ hours of intensive training in ethical hacking, threat analysis, vulnerability assessment, and incident response methodologies.
- Performed manual and automated penetration tests on web applications using Burp Suite and OWASP ZAP; identified and documented SQL injection, XSS, CSRF, and authentication bypass vulnerabilities with CVSS scoring.
- Conducted network reconnaissance and vulnerability scanning using Nmap and Nessus across lab environments; analyzed scan results to prioritize critical findings and recommend remediation strategies.
- Utilized Metasploit Framework to simulate real-world attack vectors, validate defensive controls, and support remediation planning.
- Analyzed network traffic using Wireshark to detect anomalies and potential indicators of compromise in simulated enterprise environments.

Operations & HR Intern

IKKASA | 06/2024 – 09/2024

- Managed end-to-end recruitment operations and onboarding for 15+ new hires, ensuring accurate employee records and streamlined documentation workflows.
- Coordinated cross-functional administrative activities and supported founders with operational process optimization.
- Developed standardized documentation templates that improved onboarding efficiency and reduced administrative overhead.

PROJECTS

NeuroShield NDR — AI-Driven Network Detection & Response System

Python, scikit-learn, scrapy, FastAPI, NumPy, pandas, WebSocket | github.com/349100/NeuroShield-NDR

- Architected and developed an AI-powered Network Detection and Response (NDR) system that captures network traffic, extracts 13 statistical features, and detects anomalies using hybrid machine learning (Isolation Forest + Random Forest).
- Implemented real-time packet capture and flow reassembly using scrapy with 5-tuple tracking, timeout-based expiration, and thread-safe state management for live network monitoring.
- Engineered feature extraction pipeline computing Shannon entropy, inter-arrival times, packet size distributions, TCP window sizes, and DNS query frequencies for behavioral threat detection.
- Built automated response engine with severity-based decision matrix (Critical, High, Medium, Low) mapping to quarantine, ACL block, SOC alert, and log-only actions.
- Developed FastAPI REST backend with WebSocket support for real-time dashboard updates, serving JSON threat data and live metrics to an HTML5 analytics frontend.
- Created baseline retraining script that captures real network traffic for 24+ hours and retrains the anomaly detector on user-specific normal behavior to minimize false positives.

PhantomEye — AI-Powered OSINT Intelligence Framework

Python, AsyncIO, REST APIs, DNS, Web Scraping | github.com/349100/phantomeye

- Architected a modular, asynchronous OSINT framework with plugin-based architecture supporting domain reconnaissance, social media enumeration, web intelligence, and threat intelligence modules.
- Integrated OpenAI LLM API for automated correlation analysis, pattern recognition, and natural language generation of executive security summaries.
- Built robust async HTTP client with rate limiting, retry logic, and proxy support to ensure ethical, compliant passive reconnaissance operations.
- Developed automated report generation engine outputting findings in JSON, CSV, and HTML formats with severity-based visual categorization.

SecProbe — AI-Assisted Web Security Misconfiguration Scanner

Python, Flask, REST API, Google Gemini, Groq LLaMA | github.com/349100/secprobe

- Developed an automated web security scanner that detects missing HTTP security headers, SSL misconfigurations, and exposed ports with CVSS-style risk scoring.
- Integrated free AI providers (Google Gemini, Groq LLaMA) for intelligent vulnerability explanation, impact assessment, and remediation guidance generation.
- Built Flask REST API and CLI interface supporting JSON and HTML report export with dark-themed dashboard for security assessment visualization.

Parking Management System — Security-Focused Full-Stack Application

PHP, MySQL, HTML/CSS, JavaScript, Apache

- Designed and developed a full-stack parking management system implementing secure user authentication, session management, and role-based access control.
- Mitigated SQL injection and cross-site scripting vulnerabilities through parameterized queries, input sanitization, and output encoding.

Applied secure software development lifecycle principles from requirements gathering and threat modeling to secure deployment

EDUCATION

Master of Computer Applications (MCA) — Cybersecurity & Forensic Science

Parul University, Vadodara, Gujarat | 08/2025 – 07/2027 | Expected Graduation: July 2027

Bachelor of Computer Applications (BCA)

Sambalpur University, Odisha | 08/2022 – 07/2025

CERTIFICATIONS

- Certified Ethical Hacker (CEH) v12 (Completion)
- Certified Cybersecurity Professional Program — Indian Council for Technical Research and Development.
- Cybersecurity Professional — Hacker School (A Unit of Cartel Software).

LEADERSHIP & ACTIVITIES

- Head Boy: Elected school Head Boy demonstrating leadership, organizational responsibility, and cross-functional team coordination
- Lead Graphic Designer: Led a creative design team to produce promotional materials for university events, managing timelines and deliverables